UIMS — Complete System Specification

Field	Value
Product	UIMS — University Inventory Management System
Document type	Complete System Specification (SRS + Architecture + Features + LLD)
Date	August 2026
Classification	Proprietary

Document contents

1. System Requirements Specification (SRS)
2. Architecture Design
3. Features & Requirements
4. Low-Level Design (LLD) Workflows

Part 1 — System Requirements Specification (SRS)

University Inventory Management System (UIMS)

Field	Value
Document ID	SRS-UIMS-001
Product	University Inventory Management System (UIMS)
Repository	`UIMS_SERVICE`
Version	1.0
Date	August 2026
Classification	Proprietary

---

1. Introduction

1.1 Purpose

This SRS defines requirements for **UIMS**, a university-focused inventory and property accountability platform for Philippine government institutions. It supports Supply Office workflows: master catalog, warehouse stock, PPE assets, COA-aligned forms, physical counts, GAM depreciation, requisitions, reporting, and audit compliance.

1.2 Scope

In scope

- Organization hierarchy, RBAC, item catalog, warehouses/bins, stock ledger
- Assets lifecycle, QR labels, custodianship
- COA forms: ICS, PAR, RPCPPE, IIRUP, WMR
- Physical count sessions and field PWA scanning
- GAM straight-line depreciation and property ledger
- Requisitions and supplier master
- Reports (CSV + print-to-PDF) and append-only audit log
- Local auth and University Platform Access SSO (`AUTH_MODE` local | dual | platform)
- Internal custodianship feed for UHRIS (M2M)

Out of scope / deferred

- GSIS insurance API; Bluetooth/Wi-Fi label printer and RFID sled SDKs
- Formal OCR of paper archives; dedicated procurement/DMS vendor connectors

- Formal Supply Office UAT sign-off and campus load certification
- Audit log rich web UI (API available)

1.3 Definitions

Term	Definition
PPE	Property, Plant, and Equipment
COA forms	Commission on Audit–aligned accountability documents
GAM	Government Accounting Manual depreciation practice
ICS / PAR / RPCPPE / IIRUP / WMR	Inventory Custodian Slip, Property Acknowledgement Receipt, Report on Physical Count of PPE, Inventory and Inspection Report of Unserviceable Property, Waste Material Report

1.4 References

- `UIMS_SERVICE/README.md`
- `docs/ARCHITECTURE.md`
- `docs/UIMS-Complete-Guide.md`
- `docs/PRODUCTION-READINESS.md`
- `docs/CAPACITY-AND-SECURITY.md`

---

2. Overall Description

2.1 Product Perspective

Standalone sibling of University Platform Access, UHRIS, ULMIS, and UDMS — **not embedded in UHRIS**, but may share the same PostgreSQL host under schema **`uims`**.

Surface	Port (dev)	Purpose
Staff web (`apps/web`)	5200	Supply Office SPA (optional PWA)
Field PWA	5200 `/field`	Physical count barcode scanning
API (`apps/api`)	3334	REST + OpenAPI

Health: `GET /health` → `{ status: 'ok', service: 'uims' }`

2.2 User Classes / Roles

Role	Summary
Administrator	All permissions

Supply Officer	Stock, warehouses, assets, COA, counts, depreciation, reports, audit
Property Custodian	Assets, COA, counts, depreciation read, reports
Supply Clerk	Stock write, requisitions, counts; limited catalog/warehouse
Staff	Read-oriented permissions
Viewer	Org/items/assets/reports read

Frontend permission checks are UX only; API enforces on every authenticated route.

2.3 Constraints

- Philippine government university context (COA + GAM)
- `DB_SCHEMA=uims`; separate migration history from UHRIS
- Browser never receives `SERVICE_API_KEY`
- Idempotency store requires Redis for multi-replica production
- RA 10173 applies to personal data in users, custodianships, and audits

---

3. Functional Requirements

3.1 Organization & Catalog

ID	Requirement	Priority
FR-ORG-001	Hierarchy: Organization → Campus → Office → Department	Must
FR-CAT-001	Item master: SKU, UOM, barcode, category, stock thresholds, expiry flag	Must
FR-CAT-002	Default valuation weighted average; CSV import ≤2,000 rows upsert by SKU	Must

3.2 Inventory & Requisitions

ID	Requirement	Priority
FR-INV-001	Warehouses/bins; movements: beginning, receive, issue, transfer, adjust, return, disposal, reserve, release	Must
FR-INV-002	Immutable ledger with on-hand, reserved, available balances	Must
FR-INV-003	Idempotent stock writes via `Idempotency-Key`	Must
FR-REQ-001	Requisition workflow draft → submitted → approved/rejected → fulfilled	Must

FR-REQ-002	Document numbers `REQ-{YYYYMM}-{SEQ}` ; fulfill posts stock issue	Must

3.3 Assets & COA

ID	Requirement	Priority
FR-AST-001	Asset types and lifecycle: register → assign → return → maintenance → retire → dispose	Must
FR-AST-002	QR label payload generation; custodianship assignment	Must
FR-COA-001	Forms ICS, PAR, RPCPPE, IIRUP, WMR	Must
FR-COA-002	Status draft → pending → approved | void; auto document numbers	Must
FR-COA-003	Approvals audited; export CSV; RPCPPE aligns with property counts	Must

3.4 Physical Count & Depreciation

ID	Requirement	Priority
FR-CNT-001	Count types stock and property; session open → counting → reconciling → reconciled → closed	Must
FR-CNT-002	Field PWA scan with offline queue, geo/photo evidence, sync	Must
FR-CNT-003	Reconcile posts stock adjust for variances	Must
FR-DEP-001	Straight-line GAM profiles and period entries; idempotent per asset/period	Must
FR-DEP-002	Lapsing schedule and property ledger views	Must

3.5 Audit, UHRIS Feed, Reports

ID	Requirement	Priority
FR-AUD-001	Append-only audit for stock, COA, depreciation, users, report exports	Must
FR-UHRIS-001	`GET /api/v1/internal/custodianships` protected by service API key	Must
FR-RPT-001	Reports: stock-inventory, property-ledger, consumption, depreciation	Must
FR-RPT-002	View requires `reports.read` ; export requires `reports.export` (audited)	Must

FR-DSH-001	Dashboard KPIs, near-expiry alerts, recent movements	Should

4. External Interface Requirements

Interface	Detail
REST API	`/api/v1/*` ; Swagger `/api/v1/docs`
University Platform Access SSO	Platform code `uims` ; launch `/sso/callback`
UHRIS	M2M custodianship feed; shared Postgres host optional
Object storage	S3-compatible for attachments
Observability	Optional Sentry / OpenTelemetry

5. Non-Functional Requirements

ID	Category	Requirement
NFR-CAP-001	Capacity	≥500 named users; interactive API p95 ≤3s under normal campus load
NFR-CAP-002	Capacity	Health <500 ms; support physical-count burst from field tablets
NFR-SEC-001	Security	TLS at edge; CORS locked; RBAC least privilege; non-root container
NFR-SEC-002	Privacy	RA 10173 purpose limitation, access control, retention alignment
NFR-REL-001	Reliability	Daily incremental + weekly full backups; quarterly restore drill
NFR-PERF-001	Performance	Lazy SPA routes; paginated lists; heavy jobs off request path when queued
NFR-A11Y-001	Accessibility	Skip link, table captions, aria-live where implemented

6. Verification Criteria (Summary)

Scenario	Expected result
Stock receive/issue	Ledger and balances update; idempotent retry safe
COA approve	Status approved; audit entry; property ledger impact
Physical count reconcile	Variances adjusted; session closable
Depreciation run	Entries created once per period
University Platform Access SSO	Dual mode accepts platform JWT

UHRIS custodianship	Service key required; browser cannot call

---

7. Document Control

Version	Date	Description
1.0	August 2026	SRS from UIMS architecture and readiness docs

Related design docs

- **UIMS-Architecture.md** — Architecture design (schema isolation, domains, deployment)
- **UIMS-Features-Requirements.md** — Feature catalog + requirements matrix
- **UIMS-LLD.md** — Low-level design workflows (stock, requisitions, assets, COA, physical count, depreciation, UHRIS feed)

End of SRS — UIMS

Part 2 — Architecture Design

UIMS — University Inventory Management System

Field	Value
Document ID	AD-UIMS-001
Product	UIMS
Companions	UIMS_SERVICE-SRS.md · UIMS-Features-Requirements.md · UIMS-LLD.md
Date	August 2026

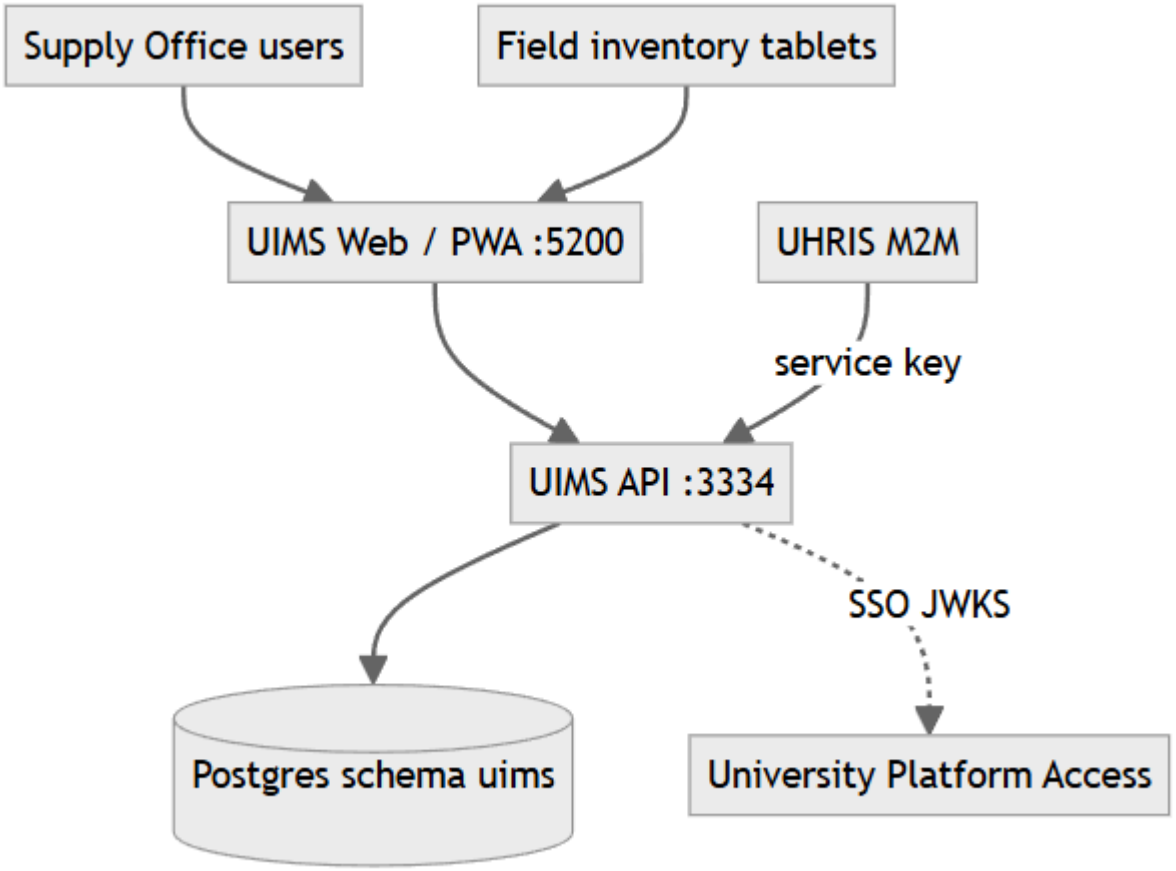
---

1. Purpose

Architecture Design for university Supply Office inventory and property accountability (COA/GAM), standalone from UHRIS but optionally sharing the same Postgres host under schema `uims`.

---

2. System context



Health: `GET /health` → `{ status: 'ok', service: 'uims' }`.

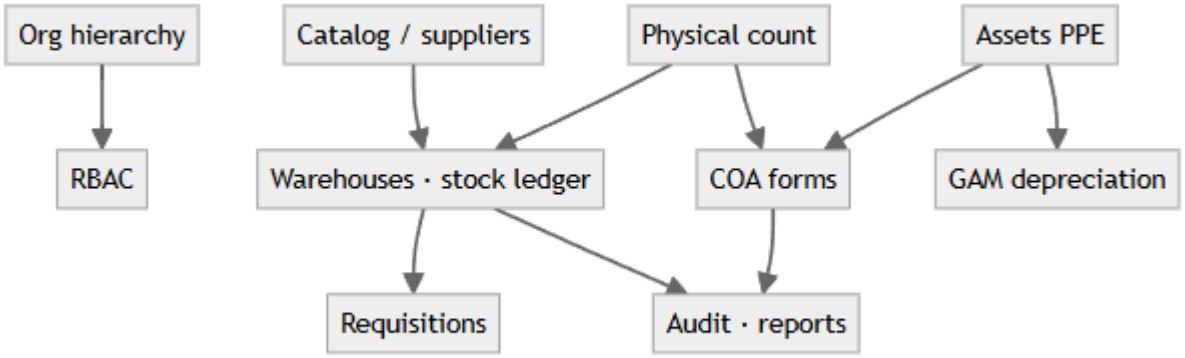
---

3. Container view

Container	Technology	Notes
Web SPA / Field PWA	React 19 + Vite	Staff UI + `/field` offline-capable scans
API	AdonisJS 7, Lucid	Port 3334
Postgres	Schema `uims` ; migration table `uims_adonis_schema`	Isolated from HRIS migrations
Auth	Local tokens and/or UPA JWKS	`AUTH_MODE` local|dual|platform

---

4. Domain module architecture



Module	Responsibility
Org	Organization → Campus → Office → Department
Catalog	Items, categories, CSV import
Inventory	Bins, movements, balances, idempotent writes
Requisitions	Request → approve → fulfill (issue)
Assets	Lifecycle, QR, custodianship
COA	ICS, PAR, RPCPPE, IIRUP, WMR
Count	Sessions + Field PWA sync
Depreciation	Straight-line profiles/entries
Internal API	Custodianships for UHRIS

5. Security architecture

Control	Detail
RBAC	27 permission codes; API enforces every route
Browser trust	UX checks only; no `SERVICE_API_KEY` in clients
CORS	Locked to production web origin
Idempotency	`Idempotency-Key` on stock writes (Redis for multi-replica)
Privacy	RA 10173 purpose limitation for custodianship/audit data

6. Deployment topology



Guidance	Detail
Sizing (500 users)	2–3 API replicas; Postgres 2–4 vCPU / 8 GB class
Backup	Daily incremental, weekly full, offsite
Observability	Optional Sentry / OpenTelemetry `uims-api`

---

7. Integration architecture

Peer	Direction	Pattern
University Platform Access	Inbound SSO	JWKS + `/sso/callback`
UHRIS	Outbound read	`GET /api/v1/internal/custodianships`
Object storage	Attachments	S3-compatible
Hardware printers/RFID	—	Out of software scope

---

8. Quality attributes

Attribute	Approach
Capacity	≥500 users; p95 ≤3s target
Integrity	Immutable stock ledger; audited COA/depreciation
Resilience	Documented RTO/RPO; quarterly restore drills
Accessibility	Skip link, table captions, aria-live

End of Architecture Design — UIMS

Part 3 — Features & Requirements

UIMS — University Inventory Management System

Field	Value
Document ID	FR-UIMS-001
Product	UIMS
Companions	UIMS_SERVICE-SRS.md · UIMS-LLD.md
Date	August 2026

---

1. Feature summary

Area	Features
Organization	Organization → Campus → Office → Department
Catalog	Items, categories, SKU/barcode, thresholds, CSV import
Inventory	Warehouses, bins, ledger movements, balances, idempotent writes
Suppliers & requisitions	Supplier master; REQ workflow to stock issue
Assets (PPE)	Register, assign, maintain, retire, QR labels
COA forms	ICS, PAR, RPCPPE, IIRUP, WMR
Physical count	Stock/property sessions + Field PWA offline scan
Depreciation	GAM straight-line profiles + period runs
Reports & audit	Stock, property ledger, consumption, depreciation; append-only audit
Integrations	University Platform Access SSO; UHRIS custodianship M2M

---

2. Feature catalog

Feature ID	Feature	Description	Priority
F-ORG-01	Org hierarchy	Org / campus / office / department	Must

F-RBAC-01	Roles & permissions	27 permission codes; seeded roles	Must
F-CAT-01	Item master	SKU, UOM, barcode, category, thresholds, expiry flag	Must
F-CAT-02	CSV import	≤2,000 rows; upsert by SKU	Must
F-INV-01	Warehouses & bins	Campus warehouses; aisle/rack/shelf bins	Must
F-INV-02	Stock movements	beginning, receive, issue, transfer, adjust, return, disposal, reserve, release	Must
F-INV-03	Ledger & balances	Immutable ledger; on-hand/reserved/available	Must
F-INV-04	Idempotency	`Idempotency-Key` on stock writes	Must
F-SUP-01	Suppliers	Supplier master + contacts	Must
F-REQ-01	Requisitions	draft → submitted → approved/rejected → fulfilled	Must
F-REQ-02	Auto numbering	`REQ-{YYYYMM}-{SEQ}`	Must
F-AST-01	PPE register	Types, statuses, lifecycle actions	Must
F-AST-02	Custodianship	Assign/return; feeds COA + UHRIS	Must
F-AST-03	QR labels	JSON payload for printing/scan	Must
F-COA-01	COA form types	ICS, PAR, RPCPPE, IIRUP, WMR	Must
F-COA-02	COA workflow	draft → pending → approved | void	Must
F-COA-03	COA export	CSV export; audited approve	Must
F-CNT-01	Physical count	stock + property sessions	Must
F-CNT-02	Field PWA	Offline queue, geo/photo, sync scans	Must
F-CNT-03	Reconcile	Variance adjust + close	Must
F-DEP-01	GAM profiles	Straight-line useful life / residual	Must
F-DEP-02	Period runs	Idempotent entries per asset/period	Must
F-RPT-01	Operational reports	stock-inventory, property-ledger, consumption, depreciation	Must
F-AUD-01	Audit log	Append-only; API list	Must
F-DSH-01	Dashboard	KPIs, near-expiry, recent movements	Should

F-AUTH-01	Local + UPA SSO	dual/platform modes	Must
F-UHRIS-01	Custodianship feed	Internal M2M for UHRIS	Must
F-CNT-04	Found-at-station	Register untagged asset discovered during Field count; create/link asset + photo/geo	Must
F-COA-04	Mobile ICS/PAR sign-off	Stylus/touch accountability signature on Field device for issued items	Should
F-DIS-01	Disposal & appraisal	Unserviceable flag → appraisal (appraised vs acquisition) → mode (bidding / donate / destroy) → WMR/IIRUP	Must
F-RPT-02	PDF layout	Export PDF with user-defined page size and orientation	Should
F-RPT-03	Valuation & lapsing	Inventory valuation + lapsing schedule (book value, accum. dep., remaining life)	Should
F-AST-04	Maintenance tracking	Repair/maintenance events on PPE	Should
F-INT-01	Procurement/finance/DMS APIs	Deferred connectors; UPA + UHRIS delivered	Could
F-MIG-01	Legacy OCR ingest	Digitization/OCR of legacy asset documents (5-year retention of history)	Could
F-INS-01	GSIS insurance tracking	Deferred insurance module for high-value PPE	Could

3. Requirements matrix

Req ID	Requirement	Related features	Priority
FR-ORG-001	Four-level org hierarchy	F-ORG-01	Must
FR-CAT-001	Item master fields	F-CAT-01	Must
FR-CAT-002	CSV import ≤2000 upsert by SKU	F-CAT-02	Must
FR-INV-001	Warehouses/bins + movement types	F-INV-01, F-INV-02	Must
FR-INV-002	Immutable ledger + balances	F-INV-03	Must
FR-INV-003	Idempotent stock writes	F-INV-04	Must
FR-REQ-001	Requisition status workflow	F-REQ-01	Must
FR-REQ-002	Numbering + fulfill issues stock	F-REQ-02, F-INV-02	Must
FR-AST-001	Asset lifecycle	F-AST-01	Must
FR-AST-002	QR + custodianship	F-AST-02, F-AST-03	Must

FR-COA-001	Five COA form types	F-COA-01	Must
FR-COA-002	Status + auto numbers	F-COA-02	Must
FR-COA-003	Approve audited + export	F-COA-03	Must
FR-CNT-001..003	Count sessions, Field PWA, reconcile	F-CNT-01..03	Must
FR-DEP-001..002	GAM profiles + idempotent runs	F-DEP-01, F-DEP-02	Must
FR-AUD-001	Append-only audit	F-AUD-01	Must
FR-UHRIS-001	Service-key custodianship API	F-UHRIS-01	Must
FR-RPT-001..002	Reports view/export permissions	F-RPT-01	Must
FR-DSH-001	Dashboard KPIs	F-DSH-01	Should
FR-CNT-004	Found-at-station untagged asset	F-CNT-04	Must
FR-COA-004	Mobile ICS/PAR digital sign-off	F-COA-04	Should
FR-DIS-001	Disposal appraisal + mode of disposal	F-DIS-01	Must
FR-RPT-003	PDF page size/orientation + valuation/lapsing	F-RPT-02, F-RPT-03	Should
FR-AST-003	Maintenance tracking	F-AST-04	Should

Source DOCX traceability (UNIVERSITY Inventory Management System.docx) — 100%

Source requirement	Feature / ID	Status
Central supplies/assets lifecycle for Supply Office	Purpose + F-*	Delivered
COA / GAM / RA 10173 compliance	F-COA-*, F-DEP-*, NFR	Delivered
Multi-campus org + RBAC	F-ORG-01, F-RBAC-01	Delivered
Item master + CSV/Excel import	F-CAT-01, F-CAT-02	Delivered
Warehouses / bins / movements / audit	F-INV-*, F-AUD-01	Delivered
Min/reorder/safety + near-expiry alerts	F-CAT-01, F-DSH-01	Delivered
Suppliers + requisitions	F-SUP-01, F-REQ-*	Delivered
Asset lifecycle + custodianship + QR	F-AST-01..03	Delivered
Repair/maintenance tracking	F-AST-04	Delivered
Physical count + Field offline PWA + geo/photo	F-CNT-01..03	Delivered
Found-at-station untagged asset	F-CNT-04	Delivered
Digital ICS/PAR sign-off on mobile	F-COA-04	Planned
COA forms ICS/PAR/RPCPPE/IIRUP/WMR	F-COA-01..03	Delivered

GAM straight-line depreciation + lapsing	F-DEP-*, F-RPT-03	Delivered
Reports stock/property/consumption/depreciation PDF/CSV	F-RPT-01, F-RPT-02	Delivered
Disposal appraisal + mode of disposal	F-DIS-01	Delivered
UPA SSO + UHRIS custodianship feed	F-AUTH-01, F-UHRIS-01	Delivered
≥500 users / ≤3s	NFR-PERF	Delivered (target)
DR daily/weekly/offsite backups	NFR-REL	Delivered
Procurement/finance/DMS connectors	F-INT-01	Excluded (deferred)
OCR of 5-year paper archives	F-MIG-01	Excluded (deferred)
GSIS insurance module	F-INS-01	Excluded (deferred)
Rugged handheld / RFID sled / Bluetooth printers	Hardware	Excluded
Training / UAT / migration / PM SLAs	SVC-UIMS-*	Service

4. Non-functional requirements (summary)

ID	Category	Requirement
NFR-CAP-001	Capacity	≥500 users; p95 ≤3s normal load
NFR-CAP-002	Capacity	Health <500 ms; count burst support
NFR-SEC-001	Security	TLS, CORS, RBAC, non-root container
NFR-SEC-002	Privacy	RA 10173 purpose limitation + retention
NFR-REL-001	Reliability	Daily/weekly backups; restore drills
NFR-PERF-001	Performance	Lazy SPA; pagination; heavy jobs off path

5. Actors & feature access

Role	Primary features
Administrator	All
Supply Officer	F-INV-*, F-REQ-*, F-AST-*, F-COA-*, F-CNT-*, F-DEP-*, F-RPT-*, F-AUD-*
Property Custodian	F-AST-*, F-COA-*, F-CNT-*, depreciation read, reports
Supply Clerk	Stock write, requisitions, counts; limited catalog
Staff / Viewer	Read-oriented org/items/assets/reports

UHRIS (M2M)	F-UHRIS-01

End of Features & Requirements — UIMS

Part 4 — Low-Level Design (LLD) Workflows

UIMS (University Inventory Management System)

Field	Value
Document ID	LLD-UIMS-001
Product	UIMS
Companion	UIMS_SERVICE-SRS.md
Date	August 2026

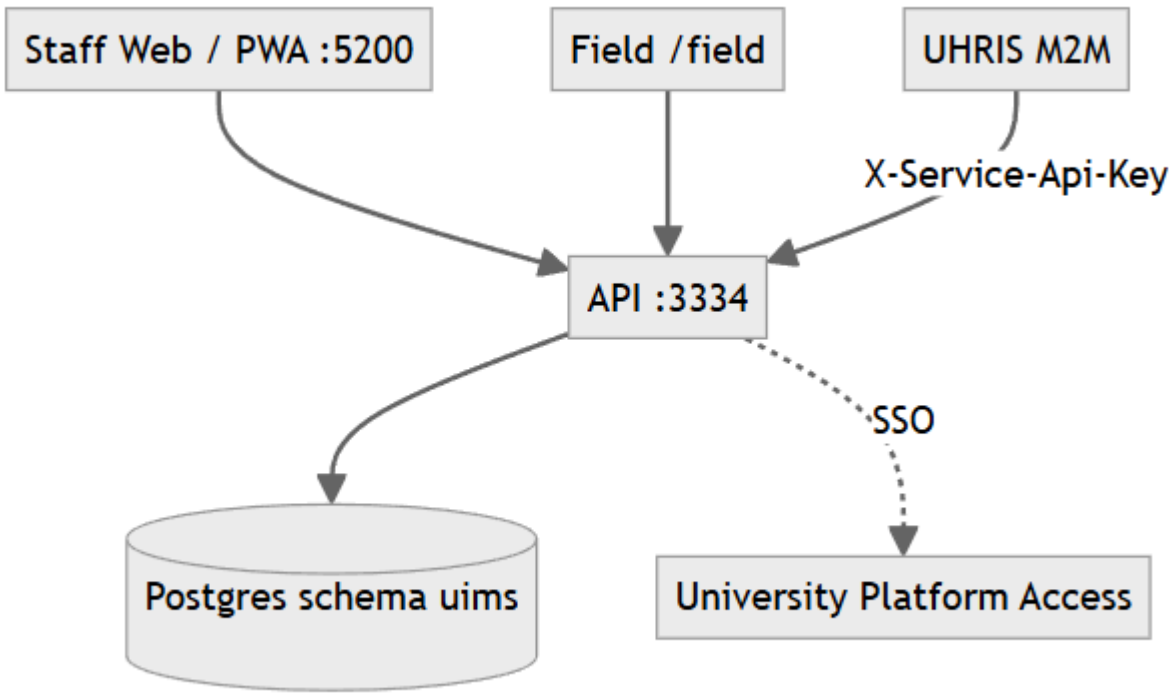
---

1. Purpose

This LLD covers stock movements, requisitions, asset custodianship, COA forms, physical count (including Field PWA), depreciation runs, and UHRIS custodianship feed.

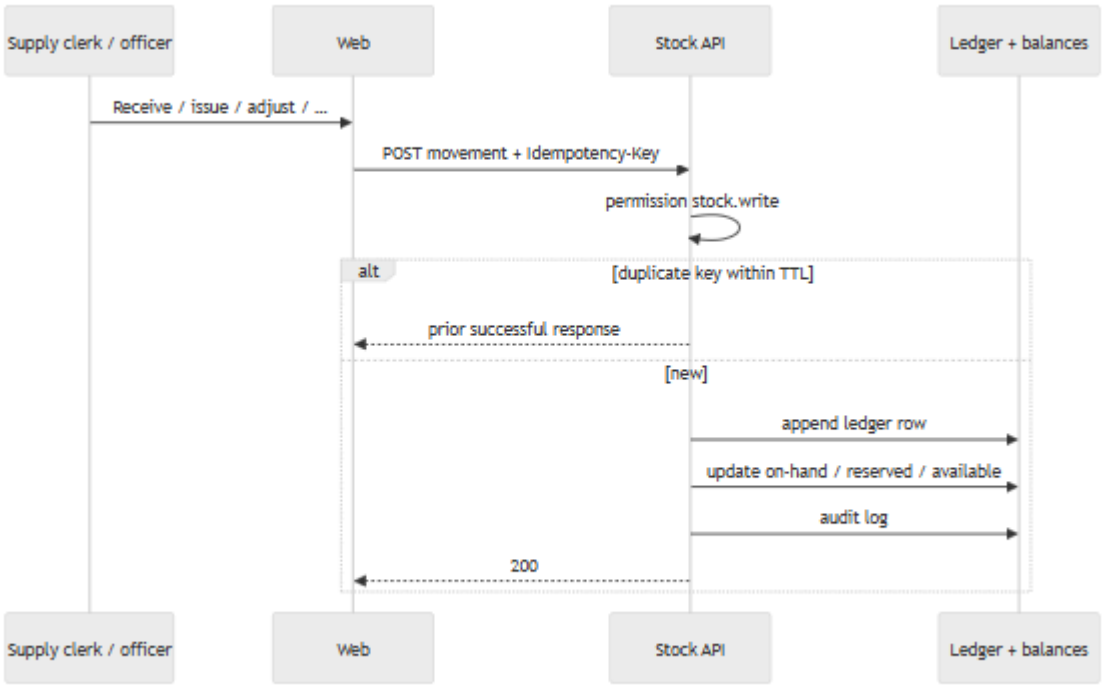
---

2. System context



---

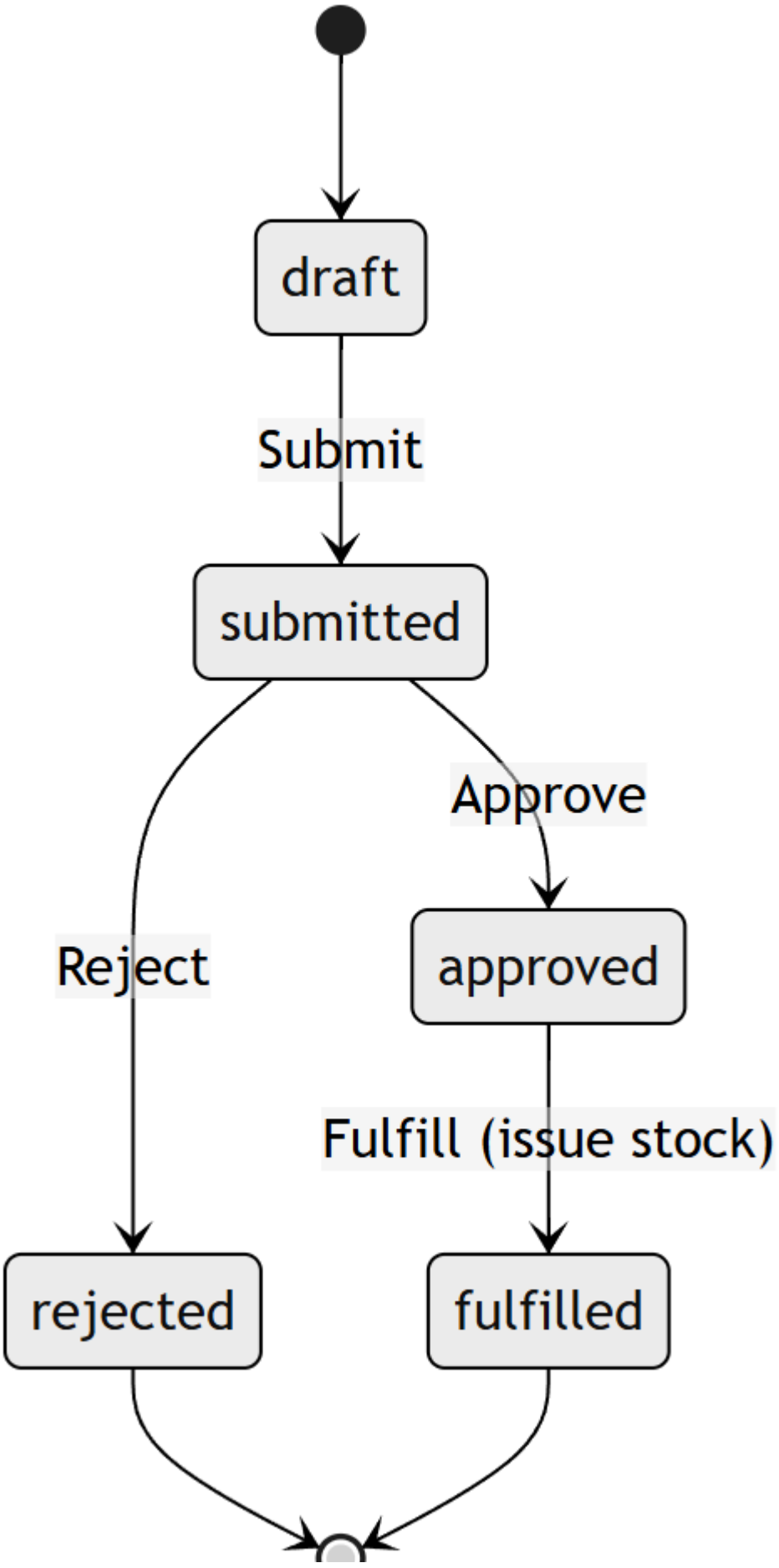
3. Stock movement workflow

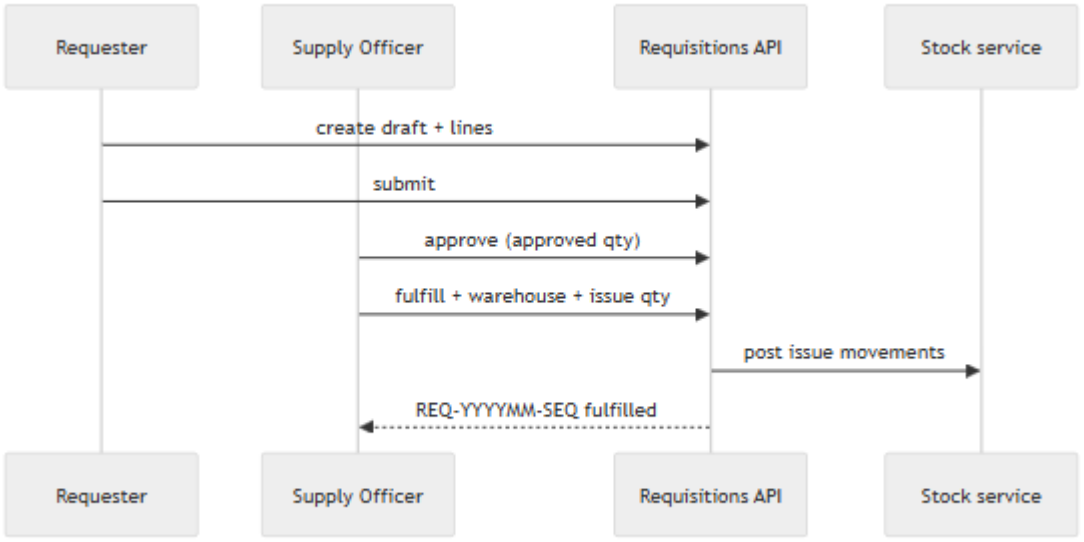


Movement types: beginning, receive, issue, transfer, adjust, return, disposal, reserve, release.

---

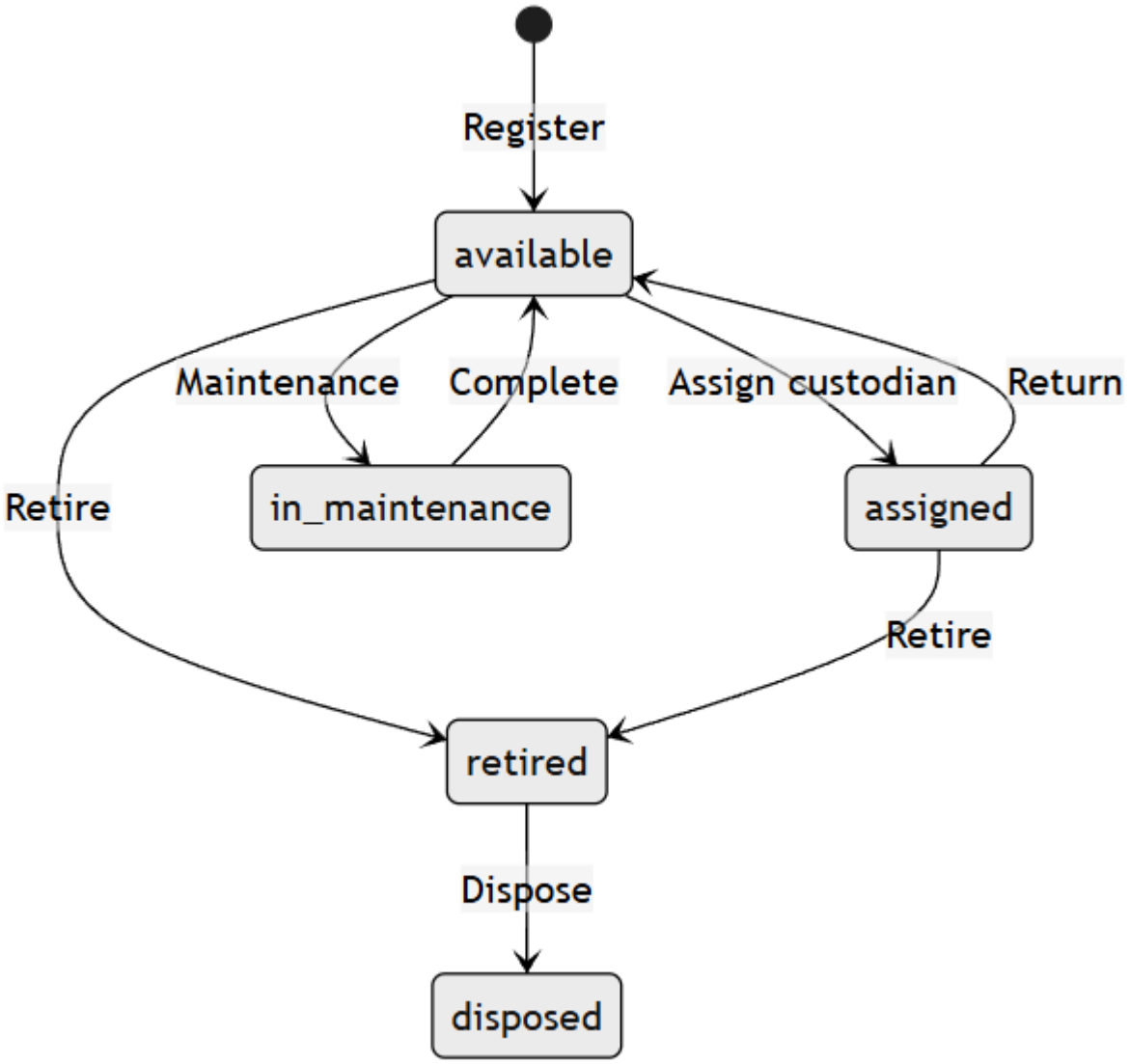
4. Requisition workflow


draft
Submit
submitted
Approve
Reject
approved
Fulfill (issue stock)
rejected
fulfilled



---

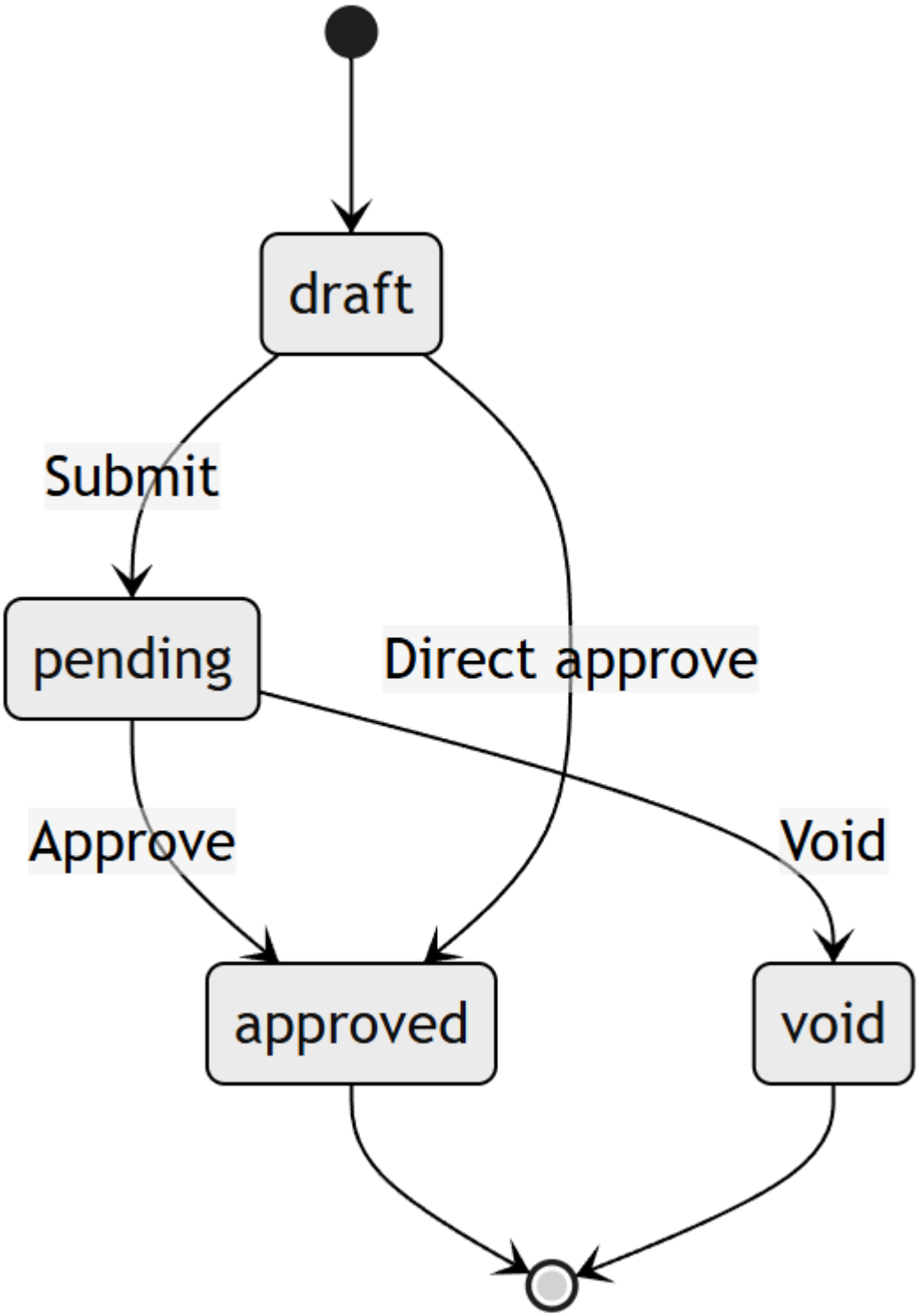
5. Asset lifecycle



QR: Generate JSON payload for label print → Field / count scan.

---

6. COA form workflow



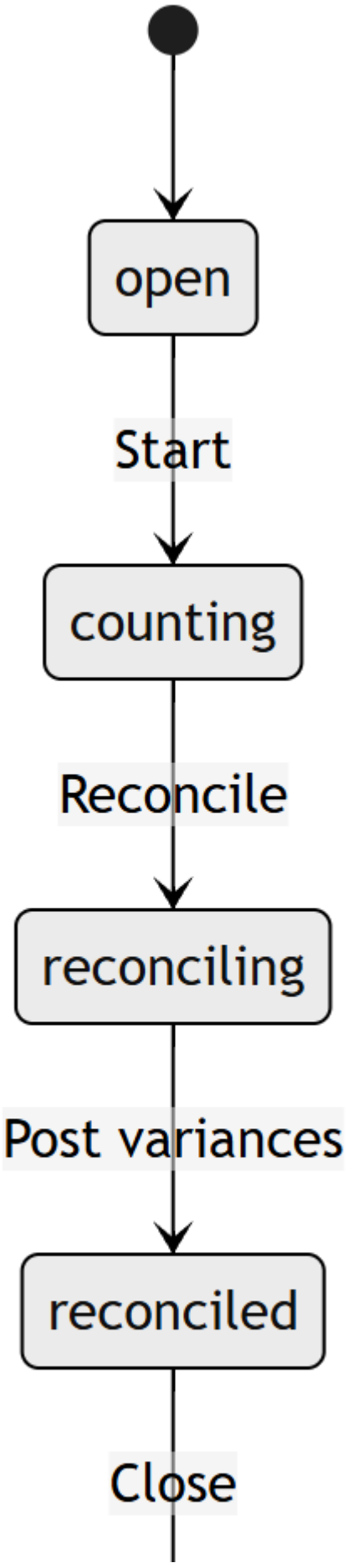
Form	Typical use
ICS	Semi-expendable to custodian
PAR	PPE acknowledgement

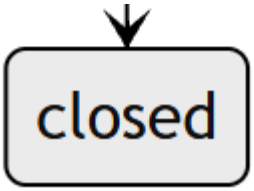
RPCPPE	Physical count of PPE
IIRUP	Unserviceable inspection
WMR	Waste material

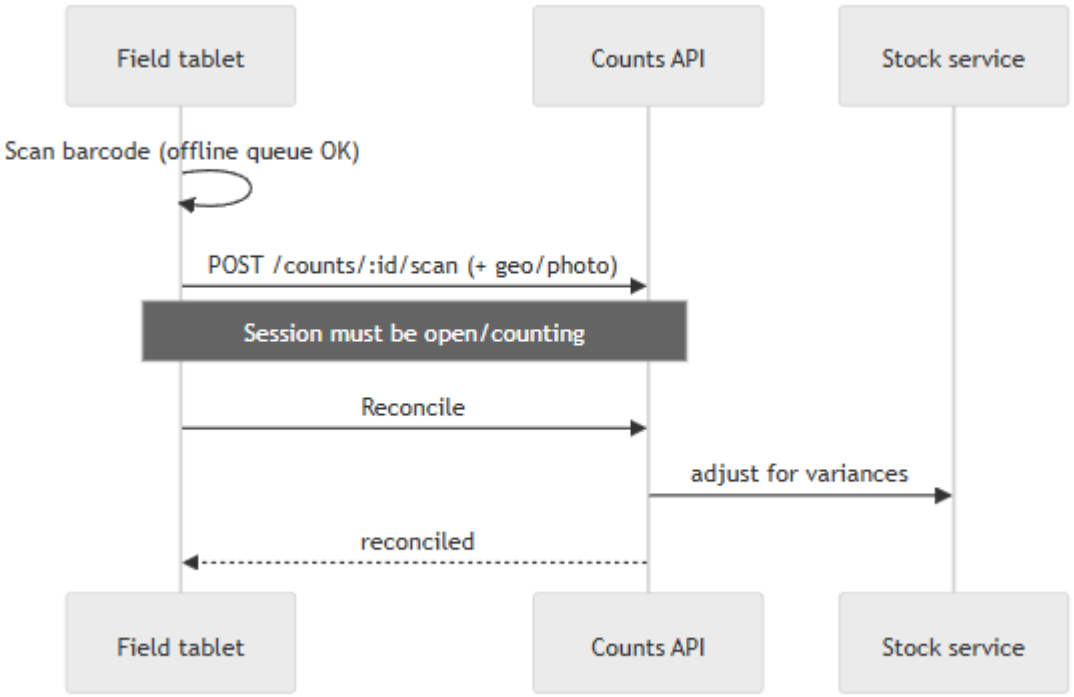
On approve: audit (`coa_approve`); property ledger entries for asset-linked forms; number `{FORM}-{YEAR}-{SEQ}` .

---

7. Physical count + Field PWA


open
Start
counting
Reconcile
reconciling
Post variances
reconciled
Close

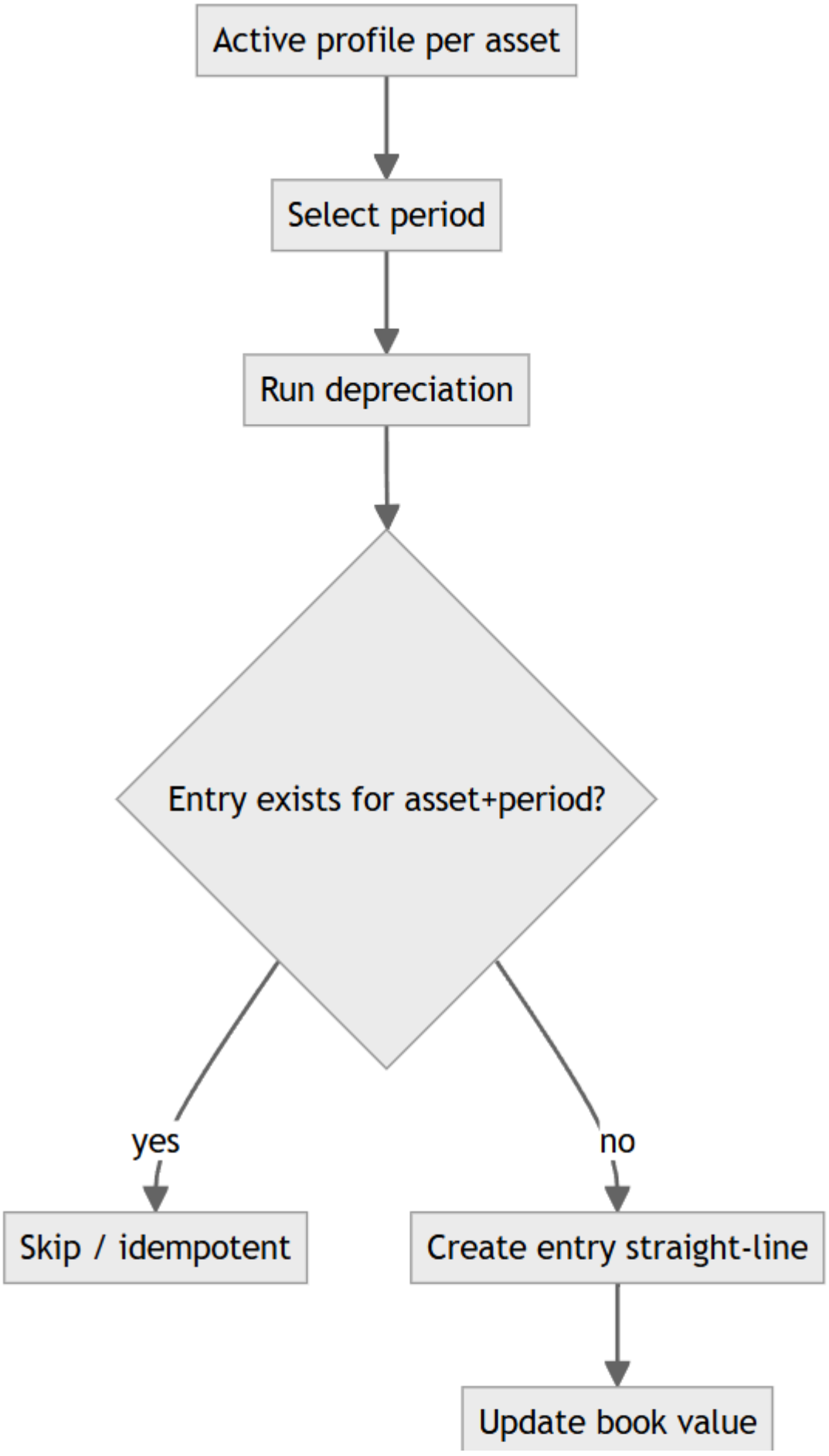


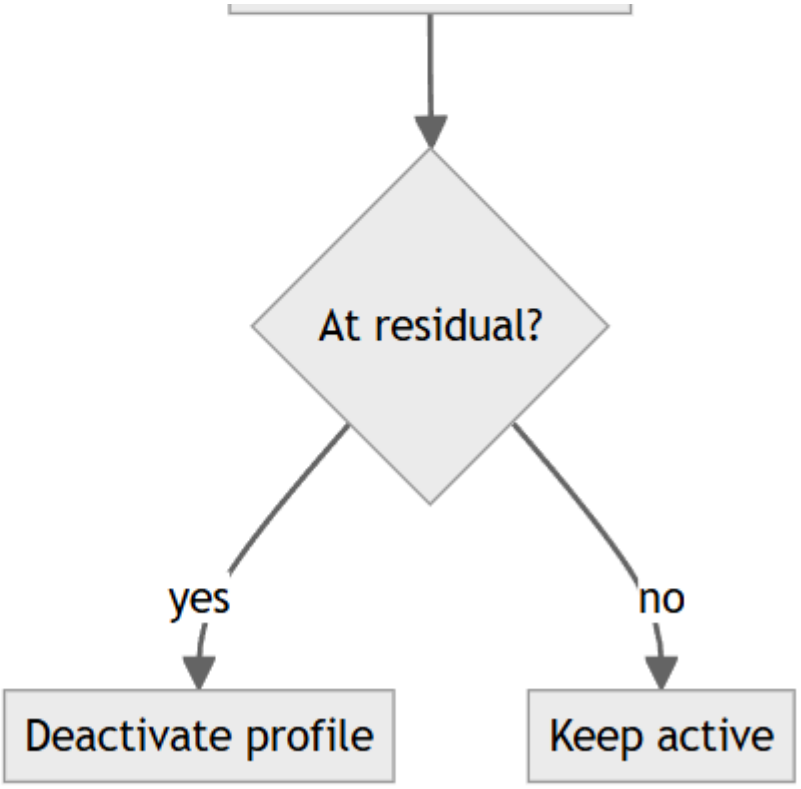


Types: `stock` (consumables) · `property` (PPE → RPCPPE alignment).

---

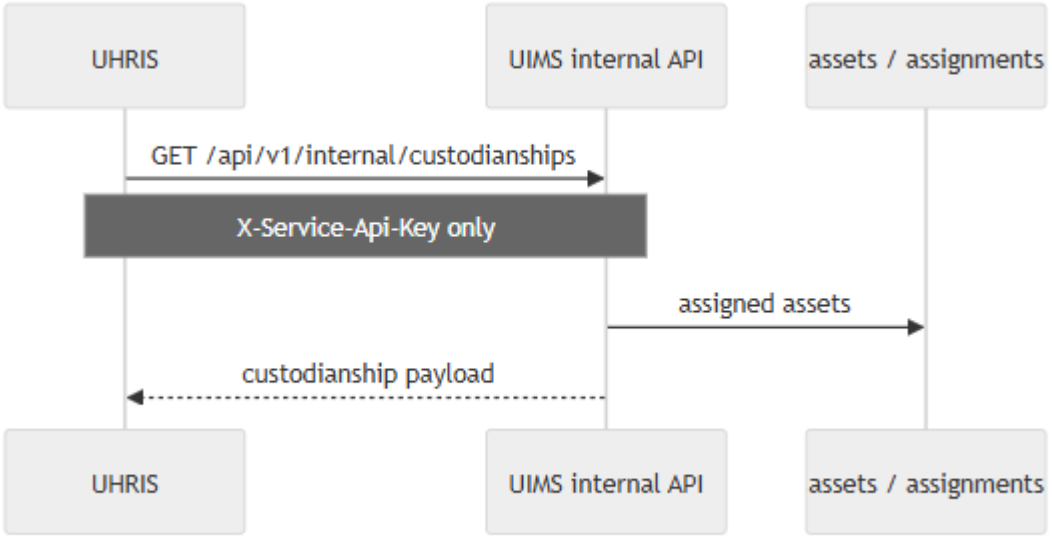
8. GAM depreciation run


Active profile per asset
Select period
Run depreciation
Entry exists for asset+period?
yes
no
Skip / idempotent
Create entry straight-line
Update book value



---

9. UHRIS custodianship feed



Browser clients must never receive `SERVICE_API_KEY`.

---

10. Error / edge cases

Case	Expected behavior
Missing Idempotency-Key on retry storm	Possible duplicate without key
Fulfill qty > approved	Rejected
Count closed	Scans rejected
COA void after approve	Use void transition where supported; audit retained
Wrong CORS origin	Browser blocked

End of LLD — UIMS
